

ODCR: Reliability-Gated Dynamic Causal Refinement for Cross-Domain Explainable Recommendation

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Abstract

Cross-domain explainable recommendation must predict target-domain preferences and generate explanations that remain content-specific, domain-appropriate, and faithful to recommendation evidence. A central failure mode is explanation degeneration, where generators learn self-reinforcing textual shortcuts instead of user–item rationale. Building on D4C, which diagnoses degeneration as a dynamic feedback process $W^{(t)} \rightarrow E^{(t)} \rightarrow W^{(t+1)}$, ODCR refines the coarse textual attribute $W^{(t)}$ into stable content evidence C , feedback-sensitive hierarchical style $S^{(t)}$, and counterfactual reliability R_{cf} . This refinement yields a complete control loop: evidence-anchored content/style decomposition, hierarchical style state modeling, reliability-gated counterfactual routing, and controlled faithful explanation generation. ODCR therefore learns explanations from $P(E \mid U, I, C, S, R_{cf})$ rather than shortcut dependence on $P(E \mid W)$. We report current results on three Amazon target-domain settings, treating them as bounded evidence for the proposed dynamic causal refinement rather than as a multi-seed benchmark-dominance claim.

1 Introduction

Explainable recommendation asks a system to predict user preferences and to justify those predictions with natural-language explanations [7]. In cross-domain settings this problem becomes especially fragile: evidence can transfer across domains, but review conventions, item vocabularies, and explanation styles can shift at the same time [8]. A generator that appears fluent in the target domain may still produce explanations that are generic, repetitive, or weakly connected to the user–item evidence. We refer to this failure mode as explanation degeneration.

D4C provides the closest causal foundation for this problem [6]. Its key insight is not only that hidden textual attributes W can act as domain-level shortcuts, but that these shortcuts participate in a dynamic feedback process. Textual attributes at time t influence explanations, and generated or observed explanations can reshape future textual attributes:

$$W^{(t)} \rightarrow E^{(t)} \rightarrow W^{(t+1)}.$$

This dynamic view explains why degeneration can be self-reinforcing. Ordinary generators can learn a shortcut from $W^{(t)}$ to $E^{(t)}$, preserving domain-looking language while weakening the genuine user–item interaction path that should support an explanation.

ODCR builds on this dynamic feedback diagnosis. Its starting point is that a single coarse textual variable $W^{(t)}$ is too broad for controlled explanation learning. The variable mixes at least three forces: stable content evidence, feedback-sensitive expression style, and the reliability of counterfactual evidence. ODCR therefore refines the D4C textual-feedback view as

$$W^{(t)} \Rightarrow (C, S^{(t)}, R_{cf}),$$

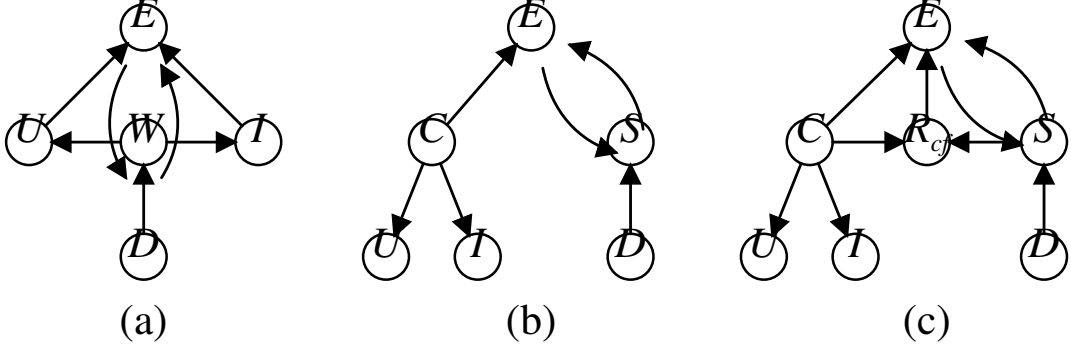


Figure 1: Dynamic causal refinement from D4C to ODCR. D4C explains explanation degeneration as a feedback loop between contextual textual attributes and explanations. ODCR refines this loop into stable content evidence, feedback-sensitive hierarchical style, and counterfactual reliability control.

where C denotes stable content evidence, $S^{(t)}$ denotes a dynamic hierarchical style state, and R_{cf} denotes the reliability with which counterfactual evidence should influence learning. The resulting method is not an engineering-stage pipeline. It is a reliability-gated dynamic causal control framework for cross-domain explainable recommendation.

Figure 1 summarizes the refinement. ODCR first anchors content/style decomposition in evidence, so that content C captures what should be explained while style $S^{(t)}$ captures how the explanation is expressed. It then models style hierarchically as a domain-global state plus instance-local residual, localizing dynamic feedback to style rather than allowing feedback to overwrite content. Counterfactual examples are further filtered and weighted by R_{cf} , so unreliable shifts do not dominate learning. Finally, the explanation generator is conditioned on (U, I, C, S, R_{cf}) , with faithfulness alignment keeping generated content consistent with rating-relevant evidence.

This manuscript makes four contributions:

- **Dynamic causal refinement.** ODCR builds on D4C’s dynamic textual-feedback diagnosis and refines coarse textual attributes into controllable variables for content, style, and counterfactual reliability.
- **Evidence-anchored decomposition.** ODCR separates stable content evidence from feedback-sensitive domain style, avoiding an undifferentiated textual shortcut.
- **Reliability-gated counterfactual control.** ODCR filters, routes, and weights counterfactual evidence using reliability and uncertainty-aware signals.
- **Controlled faithful generation.** ODCR generates explanations conditioned on content, style, and reliability-controlled evidence while keeping explanation content aligned with rating-relevant evidence.

2 Related Work

2.1 Generative Explainable Recommendation and Degeneration

Explainable recommendation studies models that pair recommendation decisions with textual or structured rationales [7]. Neural and language-model-based systems such as PETER, PEPLER,

ReXPlug, ERRa, and AdaReX demonstrate that explanations can be generated jointly with recommendation signals [3, 4, 2, 1, 5]. These models also highlight the central risk for this paper: fluent explanations are not automatically faithful explanations. When a generator overuses common review phrases or domain templates, explanation quality degenerates even if the text looks plausible.

2.2 Cross-Domain Explainable Recommendation

Cross-domain recommendation transfers evidence from a source domain to a target domain [8]. Cross-domain explainable recommendation adds a second transfer problem: the explanation must preserve recommendation content while adapting to target-domain language. This makes content and style inseparable if the model treats text as one coarse signal. ODCR addresses this by representing content evidence and style state as separate control variables rather than relying on a single domain-level textual attribute.

2.3 Causal and Counterfactual Recommendation

D4C is the closest causal foundation for ODCR [6]. It diagnoses cross-domain explanation degeneration through hidden textual attributes W and uses domain counterfactual reasoning to reduce shortcut dependence. ODCR inherits this causal view but emphasizes the dynamic feedback part of the diagnosis: $W^{(t)}$ influences explanations, and explanations can feed back into future textual attributes. ODCR refines this process into stable content evidence C , dynamic style $S^{(t)}$, and reliability-gated counterfactual influence R_{cf} .

2.4 Controlled Generation, Faithfulness, and Consistency

Controlled explanation generation requires more than producing text in the right domain style. The generated explanation should remain tied to the content evidence that supports the recommendation. ODCR therefore positions controlled generation and faithfulness alignment as output-side parts of the method: C controls what is said, $S^{(t)}$ controls how it is expressed, and R_{cf} controls which counterfactual signals are trusted during learning. Modern external systems such as CIER, MAPLE, ELIXIR, Prism, and XRec remain comparison targets once adapted to the same data and evaluation setting; the current manuscript does not claim results against them.

3 Method

3.1 Problem Formulation

Let $\mathcal{D}_s$ and $\mathcal{D}_t$ denote a source domain and a target domain. Each domain contains user-item interaction records (u, i, r, e) , where u is a user, i is an item, r is an observed rating, and e is a natural-language explanation or review-derived explanation target. Given source-domain evidence and target-domain evidence, the goal is to predict a target-domain rating $\hat{r}_{ui}$ and generate a target-domain explanation $\hat{e}_{ui}$ for a user-item pair.

The target explanation should satisfy two constraints. First, it should be content-grounded: the explanation should describe evidence relevant to the user and item rather than generic domain text. Second, it should be domain-appropriate: the explanation should use target-domain expression style without allowing style to replace recommendation evidence. ODCR formalizes these constraints through content evidence C , dynamic style $S^{(t)}$, and counterfactual reliability R_{cf} .

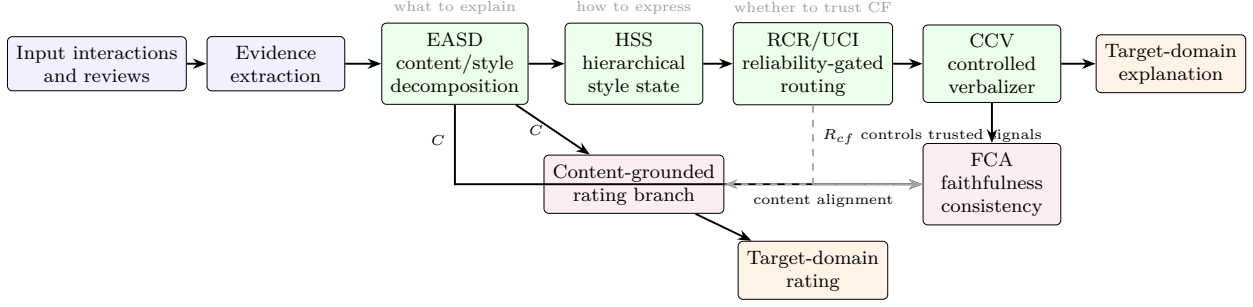


Figure 2: ODCR overall architecture. The method maps input interactions and reviews into evidence extraction, EASD content/style decomposition, HSS hierarchical style state modeling, RCR/UCI reliability-gated counterfactual routing, CCV controlled explanation generation, and FCA faithfulness consistency. The content-grounded rating branch is separated from explanation generation while sharing content evidence.

3.2 Dynamic Feedback View of Explanation Degeneration

D4C diagnoses explanation degeneration as a causal shortcut involving hidden textual attributes W [6]. In the static reading, W captures domain-level textual factors that can affect users, items, and explanations. The higher-value reading for ODCR is dynamic: textual attributes and generated explanations can reinforce one another over time,

$$W^{(t)} \rightarrow E^{(t)} \rightarrow W^{(t+1)}.$$

When this loop is left uncontrolled, a generator can learn the shortcut $W^{(t)} \rightarrow E^{(t)}$ instead of the intended interaction path $(U, I) \rightarrow E^{(t)}$. The resulting explanation may fit the target domain’s surface language while failing to explain the specific user–item preference.

This feedback view motivates a control problem. The model should preserve the part of textual evidence that supports the recommendation, allow style to adapt across domains, and prevent unreliable counterfactual shifts from being treated as trustworthy training evidence. ODCR keeps D4C’s causal diagnosis but replaces the coarse textual shortcut with explicit control variables.

3.3 From Coarse Textual Attributes to ODCR Control Variables

ODCR refines the D4C textual attribute at time t as

$$W^{(t)} \Rightarrow (C, S^{(t)}, R_{cf}).$$

Here C is stable content evidence: it captures what should be explained and what can support rating prediction. $S^{(t)}$ is a dynamic style state: it captures how content is expressed in a domain at time t . R_{cf} is counterfactual reliability: it controls whether and how strongly counterfactual evidence should influence learning.

Figure 2 shows the resulting architecture. ODCR does not ask one representation to absorb every textual factor. Instead, it uses C , $S^{(t)}$, and R_{cf} as method-level controls for content, expression, and counterfactual trust.

3.4 Evidence-Anchored Content/Style Decomposition

Evidence-Anchored Content/Style Decomposition (EASD) is the first ODCR module. It separates stable content evidence from expression style while keeping both anchored to observable user–item

evidence. Content evidence C is tied to rating-relevant aspects and interaction evidence. Style evidence captures domain phrasing, emphasis, and lexical conventions that affect how content is verbalized.

The key distinction is that ODCR does not leave shared and specific representations to infer this split without guidance. The decomposition is anchored: content is constrained by preference evidence, while style is constrained by domain-specific expression signals. This turns the coarse attribute $W^{(t)}$ into a controllable pair: C determines what should be explained, and $S^{(t)}$ determines how that evidence is expressed. Rating prediction and explanation generation can therefore share content evidence without collapsing explanation style into the rating branch.

3.5 Hierarchical Style State

Hierarchical Style State (HSS) models style as more than a domain identifier. ODCR writes the style state as

$$S^{(t)} = (S_{\text{domain}}^{(t)}, \Delta S_{\text{local}}^{(t)}),$$

where $S_{\text{domain}}^{(t)}$ captures domain-global textual conventions and $\Delta S_{\text{local}}^{(t)}$ captures instance-local residual style. The domain component models broad conventions such as review tone and domain-common phrasing. The local residual captures user-item-specific stylistic variation that should not be forced into a single domain template.

HSS makes the D4C feedback channel more precise. If explanations feed back into the future textual environment, ODCR localizes that feedback mainly to $S^{(t+1)}$. Content evidence C should remain a stable anchor for what is being recommended and explained. This prevents dynamic feedback from corrupting content while still allowing expression style to evolve with the target domain.

3.6 Reliability-Gated Counterfactual Control

Reliability-Gated Counterfactual Control combines RCR and uncertainty-aware counterfactual invariance (UCI). D4C shows that counterfactual domain evidence can reduce shortcut dependence, but ODCR further recognizes that counterfactual examples are not equally trustworthy. Some preserve content and shift only style; others alter the recommendation evidence or introduce noisy signals.

ODCR uses two control layers. The first layer is eligibility filtering: a counterfactual example is usable only if it preserves the content needed for the target explanation and does not violate the intended transfer setting. The second layer is reliability weighting and sampling priority: usable counterfactual examples receive different influence according to R_{cf} and uncertainty. A compact module-level objective is

$$\mathcal{L}_{\text{ODCR}} = \mathcal{L}_{\text{fact}} + \lambda \sum_j R_{cf,j} \mathcal{L}_{cf,j},$$

where $\mathcal{L}_{\text{fact}}$ is factual learning, $\mathcal{L}_{cf,j}$ is the counterfactual learning signal for candidate j , and $R_{cf,j}$ controls its influence. UCI reduces trust in unstable or uncertain counterfactual signals rather than treating every counterfactual example as equally informative.

Figure 3 illustrates this two-layer routing mechanism. The figure is intended as a method diagram and as an insertion point for later module ablations; it does not claim that a complete ablation suite is already integrated into the current result tables.

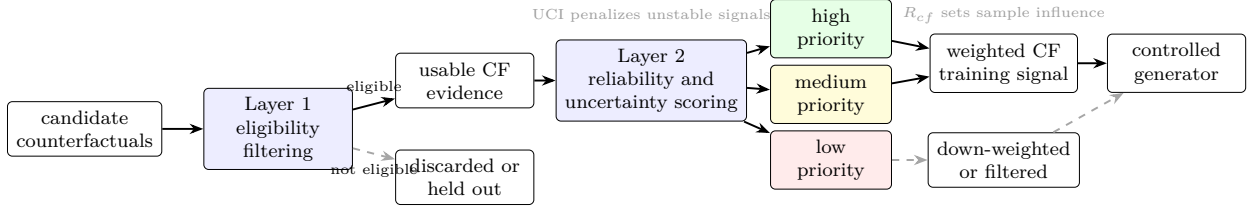


Figure 3: RCR/UCI routing mechanism. Counterfactual candidates first pass eligibility filtering, then receive reliability and uncertainty-aware weights that determine sampling priority and training influence. Low-reliability evidence is down-weighted or filtered rather than treated as equal counterfactual supervision.

3.7 Controlled Causal Verbalizer and Faithfulness Alignment

The Controlled Causal Verbalizer (CCV) is the output-side module. Instead of learning the shortcut $P(E | W)$, ODCR conditions explanation generation on the refined variables:

$$P(E | U, I, C, S, R_{cf}).$$

Content evidence C controls what should be said, style state S controls how it should be expressed, and R_{cf} controls which counterfactual training signals are trusted. This design makes explanation generation a controlled causal verbalization problem rather than an unconstrained domain-language imitation problem.

Faithfulness Consistency Alignment (FCA) keeps the generated explanation tied to content-grounded recommendation evidence. Style transfer should not substitute for preference evidence, and a fluent target-domain explanation should not drift away from the aspects that support the rating branch. FCA is therefore the consistency principle linking CCV back to C : the output may adapt its expression style, but its content should remain aligned with rating-relevant evidence.

3.8 Implementation Mapping

The implementation maps these method modules to the current ODCR workflow without making workflow stages the paper’s primary contribution. Preprocessing and Step3 support EASD/HSS by preparing evidence and learning content/style representations. Step4 supports RCR/UCI through reliability estimation, routing, confidence buckets, uncertainty signals, and sample-weight hints. Step5_e supports CCV/FCA as explanation-only controlled generation. Rating evidence remains content-grounded and separate from explanation generation, while explanation learning consumes reliability-gated evidence for target-domain verbalization.

4 Experimental Setup

4.1 Datasets and Directed Transfer Settings

The reset table system evaluates three Amazon target-domain settings. For AM-CDs, AM-Electronics and AM-Movies are used as auxiliary domains. For AM-Electronics, AM-CDs and AM-Movies are used as auxiliary domains. For AM-Movies, AM-CDs and AM-Electronics are used as auxiliary domains. In each setting, the target dataset is the evaluation dataset, and the auxiliary domain is the source of cross-domain transfer signal.

Table 1: Overall Performance on the AM-CDs Dataset

Methods	ROUGE-1	ROUGE-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	DIST-1	DIST-2	RMSE	MAE
<i>Single Domain</i>											
DualPC	12.50	9.67	9.32	3.69	1.60	0.88	7.98	0.01	0.02	0.9563	0.6736
PETER	11.62	9.07	10.42	3.86	1.77	0.99	8.41	0.78	2.64	0.9191	0.6577
ReXPlug	13.16	10.18	9.85	3.86	1.64	0.88	8.13	0.59	2.53	0.9580	0.6751
ERRA	12.72	9.92	11.53	4.32	1.99	1.12	8.88	0.42	1.46	0.9128	0.6592
<i>Cross-Domain</i>											
PEFT (E)	12.37	9.57	11.68	4.34	2.00	1.13	9.17	0.83	2.88	0.9091	0.6653
PEFT (M)	12.41	9.69	11.67	4.29	1.95	1.10	8.98	0.53	1.75	0.9050	0.6767
PEPLER (E)	12.59	9.78	11.57	4.08	1.79	0.98	8.87	0.80	3.21	0.9008	0.6561
PEPLER (M)	12.80	9.71	11.67	4.07	1.78	0.97	8.94	0.70	2.94	0.9003	0.6568
AdaReX (E)	13.42	10.37	12.64	4.54	2.00	1.08	9.48	0.67	2.62	0.9188	0.6505
AdaReX (M)	13.38	10.40	12.60	4.67	2.09	1.15	9.52	0.89	3.13	0.9178	0.6582
D4C (E)	13.50	10.61	12.71	4.69	2.23	1.21	9.69	0.93	3.57	0.9130	0.6594
D4C (M)	13.41	10.49	12.84	4.85	2.29	1.26	9.82	0.98	4.07	0.9179	0.6601
ReCEG (E)	14.02	10.98	13.06	4.98	2.40	1.34	10.12	1.04	4.28	0.9010	0.6505
ReCEG (M)	13.94	10.89	13.20	5.10	2.49	1.41	10.24	1.09	4.55	0.9030	0.6515

Note: “E” and “M” refer to AM-Electronics and AM-Movies as the auxiliary domains, respectively. Bold values indicate the best performance.

4.2 Metrics

We report automatic text-generation metrics and rating-error metrics in the same overall performance tables. The text metrics are ROUGE-1, ROUGE-L, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, DIST-1, and DIST-2. Rating errors are reported by RMSE and MAE. Larger values are better for the automatic text metrics, while smaller values are better for RMSE and MAE.

4.3 Baselines and Comparison Scope

The main tables follow the D4C-style comparison format. Single-domain baselines are DualPC, PETER, ReXPlug, and ERRA. Cross-domain baselines are PEFT, PEPLER, AdaReX, and D4C. Our method is reported as ReCEG in the result tables. D4C is the closest causal reference point for the dynamic textual-feedback view [6]. Classical explainable recommendation systems are cited as task-family context where appropriate [3, 4, 5].

4.4 Evaluation Scope

We report current results on three Amazon target-domain settings. The baseline rows follow D4C-style reporting, but standard deviations are omitted for compactness. The results should be read as bounded evidence rather than as a multi-seed benchmark claim.

5 Results

5.1 Overall Performance on AM-CDs

Table 1 reports AM-CDs as the target dataset with AM-Electronics and AM-Movies as auxiliary domains. The table uses the full D4C-style baseline set, separating single-domain methods from cross-domain methods. ReCEG improves over the corresponding D4C rows on all text-generation metrics for both auxiliary domains. The auxiliary domains show different strengths: ReCEG with AM-Electronics is stronger on ROUGE-1, ROUGE-L, and rating error among the ReCEG rows, while ReCEG with AM-Movies is stronger on BLEU, METEOR, and diversity.

Table 2: Overall Performance on the AM-Electronics Dataset

Methods	ROUGE-1	ROUGE-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	DIST-1	DIST-2	RMSE	MAE
<i>Single Domain</i>											
DualPC	10.43	8.13	9.79	2.62	0.81	0.35	7.61	0.01	0.02	1.1018	0.7876
PETER	12.68	9.55	12.04	3.87	1.44	0.65	8.89	0.64	2.00	1.0550	0.7618
ReXPlug	10.83	8.69	9.36	2.76	0.85	0.30	7.93	0.53	1.11	1.0868	0.7809
ERRA	13.27	10.01	13.25	4.30	1.56	0.69	9.55	0.64	2.04	1.0506	0.7577
<i>Cross-Domain</i>											
PEFT (C)	12.67	9.41	12.04	3.85	1.43	0.62	8.87	0.69	1.29	1.0459	0.7727
PEFT (M)	12.89	10.01	13.85	4.20	1.32	0.54	9.53	0.33	1.07	1.0508	0.7679
PEPLER (C)	14.01	10.09	12.48	3.87	1.36	0.54	10.29	0.24	1.21	1.0348	0.7657
PEPLER (M)	13.91	10.47	12.45	3.85	1.37	0.55	10.26	0.25	1.22	1.0349	0.7661
AdaReX (C)	13.27	9.98	13.11	4.25	1.53	0.66	9.60	0.50	1.63	1.0662	0.7771
AdaReX (M)	13.53	9.70	13.44	3.98	1.43	0.57	9.42	0.55	1.87	1.0622	0.7726
D4C (C)	14.02	10.42	14.16	4.39	1.69	0.78	10.32	0.68	2.21	1.0847	0.7836
D4C (M)	13.56	10.19	13.58	4.33	1.68	0.79	10.24	0.85	2.39	1.0794	0.7756
ReCEG (C)	14.38	10.68	14.50	4.58	1.82	0.88	10.65	0.76	2.54	1.0710	0.7710
ReCEG (M)	13.96	10.48	13.96	4.51	1.83	0.89	10.55	0.94	2.76	1.0660	0.7675

Note: "C" and "M" refer to AM-CDs and AM-Movies as the auxiliary domains, respectively. Bold values indicate the best performance.

5.2 Overall Performance on AM-Electronics

Table 2 reports AM-Electronics as the target dataset. ReCEG is strongest on the explanation-oriented columns: ReCEG with AM-CDs leads ROUGE-1, ROUGE-L, BLEU-1, BLEU-2, and METEOR, while ReCEG with AM-Movies leads BLEU-3, BLEU-4, DIST-1, and DIST-2. The best rating-error values still come from earlier baselines in this table, which keeps the evidence focused on explanation generation rather than recommendation prediction. This split is consistent with the paper’s content/style/reliability story: one auxiliary domain contributes closer textual relatedness, while the other contributes stronger diversity.

5.3 Overall Performance on AM-Movies

Table 3 reports AM-Movies as the target dataset. ReCEG with AM-Electronics is strongest on ROUGE and BLEU-1 through BLEU-3, matching the stronger textual relatedness visible in the D4C rows. ReCEG with AM-CDs is strongest on BLEU-4, METEOR, DIST-1, DIST-2, and MAE, preserving the higher-diversity pattern associated with the AM-CDs auxiliary domain. As in the other target-domain tables, explanation metrics show clearer movement than rating-error metrics.

5.4 Ablation Study on AM-CDs

Table 4 studies the effect of semantic evidence modeling and reliability guidance on the AM-CDs target setting. Removing both components yields the weakest ReCEG variant. Adding semantic evidence improves BLEU-3, BLEU-4, and METEOR, suggesting that evidence-aware content modeling mainly strengthens explanation quality. Adding reliability guidance improves DIST-1 and DIST-2, indicating that reliability-guided counterfactual use helps reduce repetitive generation and improves diversity. The full model achieves the best performance in both auxiliary-domain blocks, showing that semantic evidence and reliability guidance are complementary.

Table 3: Overall Performance on the AM-Movies Dataset

Methods	ROUGE-1	ROUGE-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	DIST-1	DIST-2	RMSE	MAE
<i>Single Domain</i>											
DualPC	11.83	9.10	9.21	3.04	1.23	0.63	7.94	0.03	0.06	1.0504	0.7567
PETER	14.35	10.92	13.24	4.82	2.18	1.21	9.75	0.48	1.85	0.9959	0.7411
ReXPlug	10.86	8.30	6.83	2.10	0.72	0.30	6.55	0.39	1.37	1.0236	0.7479
ERRA	13.71	10.47	12.14	4.41	1.99	1.14	9.42	0.33	1.26	1.0025	0.7413
<i>Cross-Domain</i>											
PEFT (C)	13.79	9.93	12.24	4.72	1.90	1.05	9.56	0.57	2.23	0.9944	0.7339
PEFT (E)	13.89	9.97	12.22	4.90	1.88	1.04	10.14	0.45	1.59	0.9921	0.7471
PEPLER (C)	13.66	11.20	12.80	5.25	2.20	1.29	10.21	0.59	2.98	0.9919	0.7598
PEPLER (E)	13.62	10.52	12.65	4.94	2.23	1.17	10.13	0.61	3.01	0.9920	0.7598
AdaReX (C)	13.97	10.60	12.53	4.62	2.11	1.20	9.94	0.44	2.29	0.9987	0.7305
AdaReX (E)	14.21	10.89	13.31	4.81	2.11	1.23	9.97	0.52	2.24	0.9988	0.7360
D4C (C)	14.45	11.08	13.41	4.95	2.26	1.31	10.19	1.09	4.30	1.0045	0.7352
D4C (E)	14.54	11.25	13.49	5.12	2.33	1.30	10.13	0.98	3.97	1.0071	0.7376
ReCEG (C)	14.90	11.42	13.78	5.22	2.47	1.44	10.48	1.18	4.72	0.9960	0.7290
ReCEG (E)	14.98	11.55	13.88	5.34	2.53	1.43	10.45	1.10	4.46	0.9980	0.7310

Note: "C" and "E" refer to AM-CDs and AM-Electronics as the auxiliary domains, respectively. Bold values indicate the best performance.

5.5 Discussion of the Results

Across the three tables, ReCEG consistently improves explanation metrics over D4C while retaining explicit RMSE and MAE reporting. The metric pattern is not homogeneous across auxiliary domains, and that difference is important. AM-CDs separates lexical/rating behavior from BLEU/diversity behavior across the two auxiliary domains. AM-Electronics separates lexical relevance from diversity. AM-Movies shows a parallel split, with AM-Electronics stronger on ROUGE/BLEU and AM-CDs stronger on diversity and MAE. Together with the AM-CDs ablation in Table 4, these results support the manuscript as bounded current evidence for the content/style/reliability framing, not as a state-of-the-art or statistical significance claim.

6 Discussion

6.1 Dynamic Feedback Control

ODCR treats explanation degeneration as a dynamic causal-control problem. D4C’s feedback view explains why domain textual shortcuts can become self-reinforcing: $W^{(t)}$ influences explanations, and explanations can feed back into $W^{(t+1)}$. ODCR does not discard this diagnosis. It refines the feedback loop so that stable content evidence, dynamic style, and counterfactual reliability are controlled separately.

6.2 Role of EASD and HSS

EASD and HSS are the parts of ODCR that keep the paper from becoming centered on reliability routing alone. EASD anchors content/style decomposition in evidence, separating what should be explained from how it is expressed. HSS makes style feedback hierarchical, with domain-global style and instance-local residuals. Together, they define where dynamic feedback should operate: generated explanations may reshape style, but they should not overwrite content evidence.

Table 4: Performance Comparison of Variants of Our Framework on the AM-CDs Dataset

Methods	BLEU-3	BLEU-4	METEOR	DIST-1	DIST-2
<i>AM-Electronics auxiliary domain</i>					
Original	2.23	1.21	9.69	0.93	3.57
ReCEG (w/o SE, w/o RG)	2.15	1.16	9.55	0.87	3.25
ReCEG (w/ SE, w/o RG)	<u>2.29</u>	<u>1.27</u>	<u>9.89</u>	0.90	3.43
ReCEG (w/o SE, w/ RG)	2.21	1.19	9.75	<u>0.98</u>	<u>3.82</u>
ReCEG	2.40	1.34	10.12	1.04	4.28
<i>AM-Movies auxiliary domain</i>					
Original	2.29	1.26	9.82	0.98	4.07
ReCEG (w/o SE, w/o RG)	2.20	1.18	9.64	0.90	3.70
ReCEG (w/ SE, w/o RG)	<u>2.38</u>	<u>1.34</u>	<u>10.02</u>	0.96	4.04
ReCEG (w/o SE, w/ RG)	2.28	1.25	9.88	<u>1.03</u>	<u>4.31</u>
ReCEG	2.49	1.41	10.24	1.09	4.55

Note: SE denotes Semantic Evidence modeling, and RG denotes Reliability Guidance. Underlined values indicate the best results among the three ReCEG variants excluding the full model; bold values indicate the overall best performance.

6.3 Role of RCR/UCI and CCV/FCA

RCR/UCI controls counterfactual influence after decomposition. It filters counterfactual candidates, scores reliability, and uses uncertainty to reduce trust in unstable shifts. CCV/FCA then turns the controlled evidence into target-domain explanations. CCV conditions generation on U , I , C , S , and R_{cf} ; FCA keeps explanation content aligned with content-grounded rating evidence. This closes the ODCR loop from dynamic feedback diagnosis to controlled faithful generation.

6.4 Reading the Current Evidence

The current result tables provide a compact main-text comparison over three Amazon target-domain settings, each with two auxiliary domains. The AM-CDs table shows that auxiliary domains can favor different metric families. The AM-Electronics and AM-Movies tables likewise show that auxiliary domains are not interchangeable: one domain can strengthen lexical relevance while another strengthens diversity or rating-side behavior. The AM-CDs ablation connects this pattern back to the method story. Semantic Evidence modeling supports content/evidence quality and improves BLEU-3, BLEU-4, and METEOR among the ablated variants, while Reliability Guidance supports reliability-controlled counterfactual use and improves DIST-1 and DIST-2. The full ReCEG combines both effects. These results support the manuscript as bounded current evidence; additional target-domain ablations, human assessment, multi-seed evaluation, and broader baseline adaptation can be integrated as later evidence without changing the core method story.

7 Limitations

Evaluation scope. The current manuscript reports current ReCEG values in three Amazon target-domain tables. The tables should not be interpreted as seed-robust mean-and-standard-deviation estimates or as statistical-significance evidence.

Reference and benchmark compatibility. Strict longest-reference reconstruction and full D4C paper-compatible benchmark reproduction are outside the present paper boundary. The current tables are organized for compact method presentation, with baseline values adapted from D4C-style reporting and ReCEG values shown as current bounded evidence.

Broader baselines. Modern external baselines such as CIER, MAPLE, ELIXIR, Prism, and XRec require same-setting adaptation before they can be used as result-table comparisons. They are therefore discussed as comparison scope rather than as current result claims.

Ablation scope. The current ablation table focuses on AM-CDs and on the two controls requested for the main paper story: Semantic Evidence modeling and Reliability Guidance. The manuscript therefore treats the ablation as table-supported evidence for those controls, while avoiding broader component-isolation claims that would require additional target-domain ablations.

Figure and presentation polish. The core method figures are now vector/TikZ diagrams intended for manuscript review. Their visual density, placement, and AAAI camera-ready styling may still need refinement after Chat review.

8 Conclusion

ODCR is presented as a reliability-gated dynamic causal refinement framework for cross-domain explainable recommendation. Starting from D4C’s feedback view of explanation degeneration, ODCR refines coarse textual attributes $W^{(t)}$ into stable content evidence C , feedback-sensitive hierarchical style $S^{(t)}$, and counterfactual reliability R_{cf} . This reframing turns cross-domain explanation learning into a controlled generation problem: content determines what should be explained, style determines how it is expressed, and reliability determines which counterfactual signals should shape learning.

The resulting method loop combines EASD, HSS, RCR/UCI, and CCV/FCA. It keeps rating evidence content-grounded while allowing the explanation generator to adapt target-domain language without collapsing into a coarse textual shortcut. The current tables provide bounded evidence on three Amazon target-domain settings with different auxiliary-domain behavior. Future evidence additions can extend this draft with module ablations, multi-seed evaluation, broader baseline adaptation, and human or faithfulness-oriented assessment without changing the central ODCR story.

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